

From Vectors to Bases

How a Few Vectors Can Build an Entire Space

A Simple Question

Consider the two vectors

$$v_1 = (1, 0), \quad v_2 = (0, 1).$$

Can we construct the vector

$$(3, 5)?$$

Yes:

$$(3, 5) = 3(1, 0) + 5(0, 1).$$

In fact, for any $(x, y) \in \mathbb{R}^2$,

$$(x, y) = x(1, 0) + y(0, 1).$$

Can a few vectors generate an entire space?

Linear Combinations

Given vectors $v_1, \dots, v_n$, an expression

$$a_1v_1 + \dots + a_nv_n$$

is called a **linear combination**.

For example, let

$$v_1 = (1, 2), \quad v_2 = (3, 1).$$

Then,

$$2v_1 - 3v_2 = 2(1, 2) - 3(3, 1) = (-7, 1).$$

Linear combinations are our way of building new vectors.

Span: Everything We Can Build

The **span** of $v_1, \dots, v_n$ is the set of all their linear combinations:

$$\text{span}\{v_1, \dots, v_n\} = \{a_1 v_1 + \dots + a_n v_n\}.$$

For example,

$$\text{span}\{(1, 0), (0, 1)\} = \mathbb{R}^2.$$

But,

$$\text{span}\{(1, 2)\} = \{(a, 2a) : a \in \mathbb{R}\},$$

which is just a line through the origin.

Span tells us what our vectors can generate.

But Are All the Vectors Necessary?

Consider,

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (1, 1).$$

Notice that,

$$v_3 = v_1 + v_2.$$

Therefore,

$$\text{span}\{v_1, v_2, v_3\} = \text{span}\{v_1, v_2\}.$$

The third vector is redundant.

This leads to the idea of **linear independence**.

Linear Independence

Vectors $v_1, \dots, v_n$ are **linearly independent** if

$$a_1 v_1 + \dots + a_n v_n = 0$$

implies,

$$a_1 = \dots = a_n = 0.$$

In other words:

The only way to obtain 0 is the trivial way.

For our previous example,

$$v_1 + v_2 - v_3 = 0.$$

This is a nontrivial relation, so

$$\{v_1, v_2, v_3\}$$

is linearly dependent.

Linear dependence means redundancy.

The Two Requirements

We now have two important ideas.

Spanning

The vectors are **enough** to generate the whole space.

Linear Independence

The vectors contain **no redundancy**.

So we naturally ask:

Can we have both at the same time?

Basis

A set B is a **basis** of V if

B spans V
and
 B is linearly independent.

So, informally:

A **basis** is a set of sufficient, non-redundant building blocks.

For example,

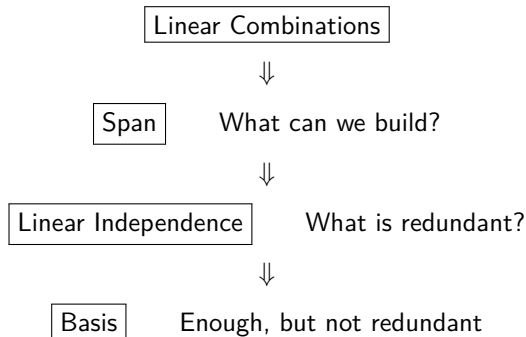
$\{(1, 0), (0, 1)\}$

is a basis of $\mathbb{R}^2$.

The Big Picture

We started with a simple question:

How can a few vectors describe an entire space?



A basis gives us a language for describing a vector space.